Jiaxin Pei

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Research Interests AI Agents, Large Language Models, Human-AI Interaction

Academic Positions

Assistant Professor, University of Texas at Austin *Aug 2026 – present*
- Department of Information, School of Computing
- Affiliated with UT Austin NLP Group and the Oden Institute for Computational Engineering and Sciences

Research Scientist, Stanford University *2025 – 2026*
- Stanford Institute for Human-Centered Artificial Intelligence (HAI)

Postdoctoral Scholar, Stanford University *2024 – 2025*
- Stanford Institute for Human-Centered Artificial Intelligence (HAI)
- Stanford Digital Economy Lab and Stanford NLP Group
- Advisors: Alex "Sandy" Pentland, Diyi Yang, and Erik Brynjolfsson

Education

University of Michigan 2019 - 2024
- Ph.D. in Information Science
- Advisor: David Jurgens

Wuhan University 2015 - 2019
- B.E. in Computer Science (Honors program)

Other Positions

Bloomberg AI 2023
- *Research Intern, Knowledge and Information Extraction Team*

Snapchat Research 2022
- *Research Intern, Computational Social Science Team*

Nokia Bell Labs 2020
- *Research Intern, Social Dynamics Group*

Awards and Honors

Best Demo Paper Award, the 12th AAAI Conference on Human Computation and Crowdsourcing (HCOMP-2024)

Best Student Paper Award, the 3rd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO-2023)

Honorable Mention Award, the 9th International Conference on Computational Social Science (IC2S2-2023)

Best Paper Award, the 1st Workshop on Social Influence in Conversations @ACL 2023

Grants

Gates Foundation Research Grant ($375,000) 2026

Teaching

INF 385T: Building Large Language Model Applications Fall 2026

Advising & Mentoring

PhD Students, University of Texas at Austin
- Shenzhe Zhu (Information Studies)
- Jingyu Tang (Information Studies)
- Keyang Xuan (Computer Science, co-advised with Yizhong Wang)

Master's Students and Research Interns
- Yuxin Chen (Master's in CS, UT Austin)
- Xu Yang (Master's in CS, UT Austin)
- Andy Dai (Master's in CS, Stanford)
- Hannah Guan (Master's in Symbolic Systems, Stanford)
- Xiangning Lin (Research Intern, Stanford HAI)

Past Mentees *(next position)*
- Zhenyu Zhang (Master's in CS, Stanford) → Member of Technical Staff, OpenAI
- Zhemin Huang (Master's in CS, Stanford) → Member of Technical Staff, Microsoft AI
- Anshika Agarwal (Master's in CS, Stanford) → Meta Superintelligence Labs
- Tim Chen (Master's in CS, Stanford) → Software Engineer, Stripe
- Mingqian Zheng (Master's in DS, University of Michigan) → PhD, Carnegie Mellon University
- Huaman Sun (Master's in DS, University of Michigan) → PhD, Northwestern University

Working Papers

[12] Xiangning Lin, Shenzhe Zhu, Shu Yang, Zhenyu Zhang, Haoqian Zhang, Yipeng Zhao, Chengxuan Qian, Tianwei Wang, Ziheng Zhang, Zhenlong Yuan, Dingcheng Wang, Juncheng Wu, Yuan Si, Jiaxin Liu, Baolong Bi, Robert Mahari, Tobin South, Dazza Greenwood, Zexue He, Rishi Bommasani, Sophia Kazinnik, Andreas Haupt, Samuele Marro, Erik Brynjolfsson, Alex Pentland, **Jiaxin Pei**. *AISPA: User-Centric System Prompt Auditing for Large Language Model Applications.* Under review.

[11] Inyoung Cheong, Wenyue Hua, Robert Mahari, Tobin South, Zexue He, Alex Pentland, **Jiaxin Pei**. *AI Agents Are Not Ready to be Agents: Loyalty, Trust, and Accountability Issues in AI Delegation.* Under review.

[10] Yuxiang Lin, Zihan Wang, Mengyang Liu, Yuxuan Shan, Longju Bai, Junyao Zhang, Xing Jin, Boshan Chen, Jinyan Su, Xingyao Wang, **Jiaxin Pei**, Manling Li. *BAGEN: Are LLM Agents Budget-Aware?* Under review.

[9] Longju Bai, Zhemin Huang, Xingyao Wang, Jiao Sun, Rada Mihalcea, Erik Brynjolfsson, Alex Pentland, **Jiaxin Pei**. *How Do AI Agents Spend Your Money? Analyzing and Predicting Token Consumption in Agentic Coding Tasks.* Under review.

[8] Shu Yang, Shenzhe Zhu, Hao Zhu, José Ramón Enríquez, Di Wang, Alex Pentland, Michiel A. Bakker, **Jiaxin Pei**. *Multi-User Large Language Model Agents.* Under review.

[7] Shenzhe Zhu, Jiao Sun, Yi Nian, Tobin South, Alex Pentland, **Jiaxin Pei**. *The Automated but Risky Game: Modeling Agent-to-Agent Negotiations and Transactions in Consumer Markets.* Under review.

[6] Sophia Kazinnik, José Ramón Enríquez, Jacy Anthis, David Nguyen, **Jiaxin Pei**. *Sex, Drugs, and LLMs: Social Desirability Bias in Language Models.* Under review.

[5] Keyang Xuan, Peiyang Song, Pan Lu, Pengrui Han, Wenkai Li, Zhenyu Zhang, Zexue He, Wenyue Hua, Manling Li, Jiaxuan You, Adrian Weller, Yizhong Wang, **Jiaxin Pei**. *Interactive Evaluation Requires a Design Science.* Under review.

[4] Yijia Shao, Humishka Zope, Yucheng Jiang, **Jiaxin Pei**, David Nguyen, Erik Brynjolfsson, Diyi Yang. *Future of Work with AI Agents: Auditing Automation and Augmentation Potential across the U.S. Workforce.* Under review.

[3] **Jiaxin Pei**, José Ramón Enríquez, Umar Patel, Alia Braley, Nuole Chen, Lily Tsai, and Alex Pentland. *Deliberation.io: Facilitating Democratic and Civil Engagement at Scale with Open-Source and Open-Science.* Under review.

[2] Samuele Marro, Alan Chan, Xinxing Ren, Lewis Hammond, Jesse Wright, Gurjyot Wanga, Tiziano Piccardi, Nuno Campos, Tobin South, Jialin Yu, Alex Pentland, Philip Torr, **Jiaxin Pei**. *Permission Manifests for Web Agents.* Under review.

[1] Tobin South, Subramanya Nagabhushanaradhya, Ayesha Dissanayaka, Sarah Cecchetti, George Fletcher, Victor Lu, Aldo Pietropaolo, Dean H Saxe, Jeff Lombardo, Abhishek Maligehalli Shivalingaiah, Stan Bounev, Alex Keisner, Andor Kesselman, Zack Proser, Ginny Fahs, Andrew Bunyea, Ben Moskowitz, Atul Tulshibagwale, Dazza Greenwood, **Jiaxin Pei**, Alex Pentland. *Identity Management for Agentic AI: The New Frontier of Authorization, Authentication, and Security for an AI Agent World.* Under review.

Peer-reviewed Proceedings

[32] Jiachen Liu, **Jiaxin Pei**, Jintao Huang, Chenglei Si, Ao Qu, Xiangru Tang, Runyu Lu, Lichang Chen, Xiaoyan Bai, Haizhong Zheng, Carl Chen, Zhiyang Chen, Haojie Ye, Yujuan Fu, Zexue He, Zijian Jin, Zhenyu Zhang, Shangquan Sun, Maestro Harmon, John Dianzhuo Wang, Jianqiao Zeng, Jiachen Sun, Mingyuan Wu, Baoyu Zhou, Chenyu You, Shijian Lu, Yiming Qiu, Fan Lai, Yuan Yuan, Yao Li, Junyuan Hong, Ruihao Zhu, Beidi Chen, Alex Pentland, Ang Chen, Mosharaf Chowdhury, Zechen Zhang. *The Last Human-Written Paper: Agent-Native Research Artifacts.* Proceedings of the 2026 Conference on Neural Information Processing Systems (NeurIPS'26).

[31] *__Jiaxin Pei__, *Zhihan Wang, and Jun Li. *70 Million Canvas Grading Records Reveal Widespread Sequential Bias and System-Induced Surname Initial Disparity.* Management Science (forthcoming). **Best Student Paper Award at EAAMO'23.**

[30] Zexue He, Yu Wang, Churan Zhi, Yuanzhe Hu, Tzu-Ping Chen, Lang Yin, Ze Chen, Tong Arthur Wu, Siru Ouyang, Zihan Wang, **Jiaxin Pei**, Julian McAuley, Yejin Choi, Alex Pentland. *MemoryArena: Benchmarking Agent Memory in Interdependent Multi-Session Agentic Tasks.* Proceedings of the 2026 International Conference on Machine Learning (ICML'26).

[29] Anand V. Shah, Tobin South, Talfan Evans, Hannah Rose Kirk, **Jiaxin Pei**, Andrew Trask, E. Glen Weyl, Michiel A. Bakker. *Position: Robust AI Personalization Will Require a Human Context Protocol.* Proceedings of the 2026 International Conference on Machine Learning (ICML'26, Position Paper).

[28] José Ramón Enríquez, **Jiaxin Pei**, Alex Pentland. *Position: AI Should Facilitate Democratic Deliberation at Scale.* Proceedings of the 2026 International Conference on Machine Learning (ICML'26, Position Paper, Spotlight).

[27] Joachim Baumann, **Jiaxin Pei**, Sanmi Koyejo, Dirk Hovy. *Position: Stop Automating Peer Review Without Rigorous Evaluation.* Proceedings of the 2026 International Conference on Machine Learning (ICML'26, Position Paper, Spotlight).

[26] Shenzhe Zhu, Shu Yang, Michiel A. Bakker, Alex Pentland, **Jiaxin Pei**. *Can AI Truly Represent Your Voice in Deliberations? A Comprehensive Study of Large-Scale Opinion Aggregation with LLMs.* Proceedings of the 2026 Conference on Language Modeling (COLM'26).

[25] Arpandeep Khatua, Hao Zhu, Peter Tran, Arya Prabhudesai, Frederic Sadrieh, Johann K. Lieberwirth, Xinkai Yu, Yicheng Fu, Michael J. Ryan, **Jiaxin Pei**, Diyi Yang. *CooperBench: Why Coding Agents Cannot be Your Teammates Yet.* Proceedings of the 2026 Conference on Language Modeling (COLM'26).

[24] Jing Yang, Qiyao Wei, **Jiaxin Pei**. *Paper Copilot: Tracking the Evolution of Peer Review in AI Conferences.* Proceedings of the 2026 International Conference on Learning Representations (ICLR'26).

[23] Elinor Poole-Dayan, Jiayi Wu, Taylor Sorensen, **Jiaxin Pei**, Michiel A. Bakker. *Benchmarking Overton Pluralism in LLMs.* Proceedings of the 2026 International Conference on Learning Representations (ICLR'26).

[22] Zhenyu Zhang, Tianyi Chen, Weiran Xu, Alex Pentland, and **Jiaxin Pei**. *ReCAP: Recursive Context-Aware Reasoning and Planning for Large Language Model Agents.* Proceedings of the 2025 Conference on Neural Information Processing Systems (NeurIPS'25).

[21] Harsh Nishant Lalai, Raj Sanjay Shah, **Jiaxin Pei**, Sashank Varma, Yi-Chia Wang, and Ali Emami. *The World According to LLMs: How Geographic Origin Influences LLMs' Entity Deduction Capabilities.* Proceedings of the 2025 Conference on Language Modeling (COLM'25).

[20] Matthias Orlikowski, **Jiaxin Pei**, Paul Röttger, Philipp Cimiano, David Jurgens, and Dirk Hovy. *Beyond Demographics: Fine-tuning Large Language Models to Predict Individuals' Subjective Text Perceptions.* Proceedings of the 2025 Annual Meeting of the Association for Computational Linguistics (ACL'25).

[19] †Huaman Sun, **Jiaxin Pei**, Minje Choi and David Jurgens. *Sociodemographic Prompting is Not Yet an Effective Approach for Simulating Subjective Judgments with LLMs.* Proceedings of the 2025 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL'25).

[18] Yinuo Xu, Hong Chen, Sushrita Rakshit, Aparna Ananthasubramaniam, Omkar Yadav, Mingqian Zheng, Michael Jiang, Lechen Zhang, Bowen Yi, Kenan Alkiek, Abraham Israeli, Bangzhao Shu, Hua Shen, **Jiaxin Pei**, Haotian Zhang, Miriam Schirmer, and David Jurgens. *Causally Modeling the Linguistic and Social Factors that Predict Email Response.* Proceedings of the 2025 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL'25).

[17] Xianzhe Fan, Qing Xiao, Xuhui Zhou, **Jiaxin Pei**, Maarten Sap, Zhicong Lu, and Hong Shen. *User-Driven Value Alignment: Understanding Users' Perceptions and Strategies for Addressing Biased and Discriminatory Statements in AI Companions.* The 2025 ACM CHI Conference on Human Factors in Computing Systems (CHI'25).

[16] †Ruijing Zhao, †Brian Diep, **Jiaxin Pei**, Dongwook Yoon, David Jurgens, and Jian Zhu. *Who Reaps All the Superchats? A Large-Scale Analysis of Income Inequality in Virtual YouTuber Livestreaming.* The 2025 ACM CHI Conference on Human Factors in Computing Systems (CHI'25).

[15] **Jiaxin Pei**, Soumya Vadlamannati, Liang-Kang Huang, Daniel Preotiuc-Pietro and Xinyu Hua. *Modeling and Detecting Company Risks from News.* Proceedings of the 2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL'24, Industry Track).

[14] †Mingqian Zheng, **Jiaxin Pei**, Lajanugen Logeswaran, Moontae Lee and David Jurgens. *When "A Helpful Assistant" Is Not Really Helpful: Personas in System Prompts Do Not Improve Performances of Large Language Models.* Findings of the 2024 Conference on Empirical Methods in Natural Language Processing (Findings of EMNLP'24).

[13] ***Jiaxin Pei**, *Minje Choi, Sagar Kumar, Chang Shu and David Jurgens. *Do LLMs Understand Social Knowledge? Evaluating the Sociability of Large Language Models with SocKET Benchmark.* Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP'23).

[12] Dimosthenis Antypas, Asahi Ushio, Francesco Barbieri, Leonardo Neves, Kiamehr Rezaee, Luis Espinosa-Anke, **Jiaxin Pei**, Jose Camacho-Collados. *SuperTweetEval: A Challenging, Unified and Heterogeneous Benchmark for Social Media NLP Research.* Findings of the 2023 Conference on Empirical Methods in Natural Language Processing (Findings of EMNLP'23).

[11] **Jiaxin Pei** and David Jurgens. *When Do Annotator Demographics Matter? Measuring The Influence of Annotator Demographics with the POPQUORN Dataset.* Proceedings of the 17th Linguistic Annotation Workshop (LAW-XVII) @ACL 2023.

[10] Aparna Ananthasubramaniam, Hong Chen, Jason Yan, Kenan Alkiek, **Jiaxin Pei**, Agrima Seth, Lavinia Dunagan, Minje Choi, Benjamin Litterer, and David Jurgens. *Exploring Linguistic Style Matching in Online Communities: The Role of Social Context and Conversation Dynamics.* Proceedings of the 1st Workshop on Social Influence in Conversations (SICon) @ACL 2023. **Best Paper Award.**

[9] **Jiaxin Pei**, Vítor Silva, Maarten Bos, Yozen Liu, Leonardo Neves, David Jurgens and Francesco Barbieri. *SemEval 2023 Task 9: Multilingual Tweet Intimacy Analysis.* Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023).

[8] **Jiaxin Pei**, Aparna Ananthasubramaniam, Xingyao Wang, Naitian Zhou, Jackson Sargent, Apostolos Dedeloudis and David Jurgens. *POTATO: The Portable Text Annotation Tool.* Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP'22 Demo). **Best Demo Paper Award at HCOMP'24.**

[7] ***Jiaxin Pei**, *Dustin Wright, David Jurgens and Isabelle Augenstein. *Modeling Information Change in Science Communication with Semantically Matched Paraphrases.* Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP'22). **Honorable Mention Award at IC2S2'23.**

[6] **Jiaxin Pei** and David Jurgens. *Measuring Sentence-level and Aspect-level (Un)certainty in Science Communications.* Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP'21).

[5] **Jiaxin Pei** and David Jurgens. *Quantifying Intimacy in Language.* Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP'20).

[4] ***Jiaxin Pei**, *Canwen Xu, Hongtao Wu, Yiyu Liu and Chenliang Li. *MATINF: A Jointly Labeled Large-Scale Dataset for Classification, Question Answering and Summarization.* Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL'20).

[3] Yu Duan, Canwen Xu, **Jiaxin Pei**, Jialong Han and Chenliang Li. *Pre-train and Plug-in: Flexible Conditional Text Generation with Variational Auto-Encoders.* Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL'20).

[2] Canwen Xu, Jing Li, Xiangyang Luo, **Jiaxin Pei**, Chenliang Li, Donghong Ji. *DLocRL: A Deep Learning Pipeline for Fine-Grained Location Recognition and Linking in Tweets.* Proceedings of the Web Conference 2019 (WWW'19).

[1] **Jiaxin Pei** and Chenliang Li. *S2SPMN: A Simple and Effective Framework for Response Generation with Relevant Information.* Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP'18).

Past Grants

UBC Language Science Research Seed Funds ($5,000), with Jian Zhu — 2024

Rackham Graduate Student Research Grant ($1,500) — 2024

Prolific Academic Research Gift ($10,000), with David Jurgens — 2023

Rackham Graduate Student Research Grant ($1,500) — 2021

Rackham Conference Travel Grant — 2021 - 2024

UMSI Conference Travel Grant — 2021 - 2024

Service

Conference Area Chair: ACL Rolling Review, NeurIPS, CHI, ICLR, COLM

Conference PC/Reviewer: ICML 2026, AAAI 2020-2022, IJCAI 2023, ACL Rolling Review 2021-2026, IC2S2 2022-2024, ACL 2021-2023, EMNLP 2021-2023, EACL 2023, NLPCC 2021-2022, ICIS 2023, NLP&CSS Workshop 2022, SemEval 2023

Journal Reviewer: Journal of the Association for Information Science and Technology (JASIST), Nature Human Behavior, Proceedings of the National Academy of Sciences of the United States of America (PNAS)